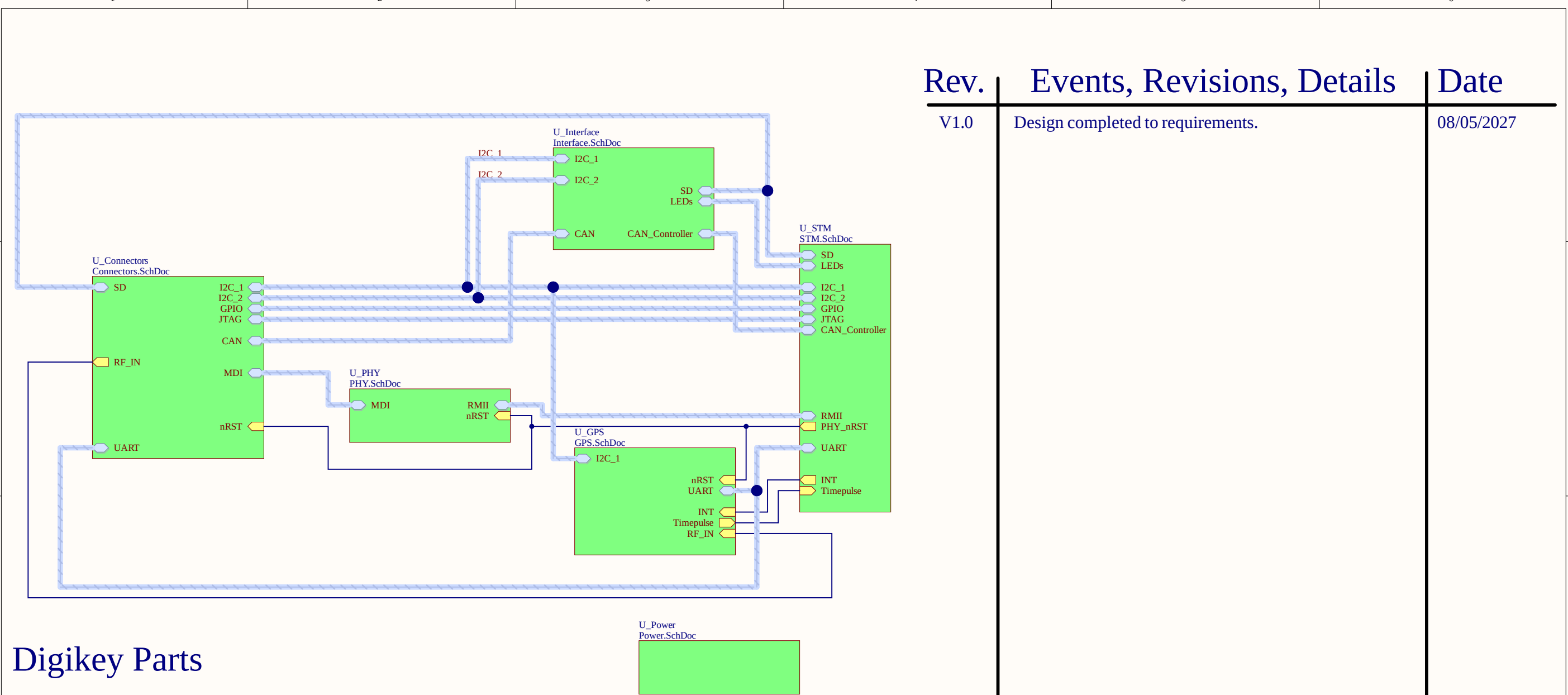
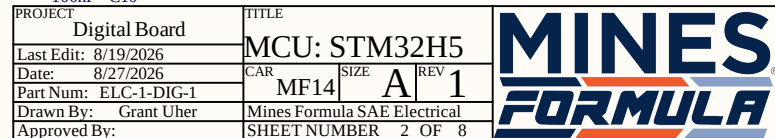


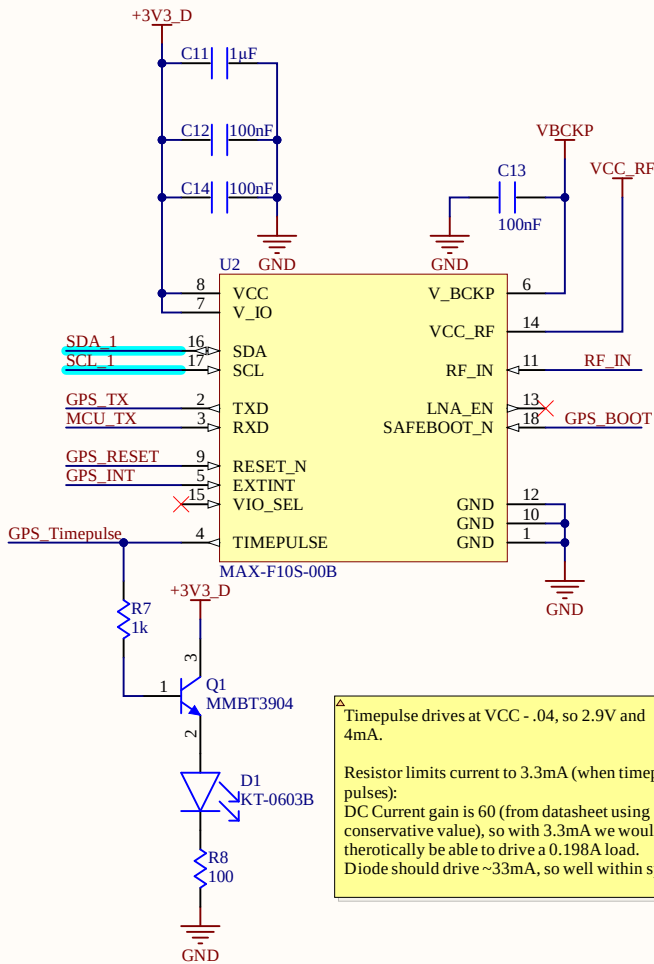
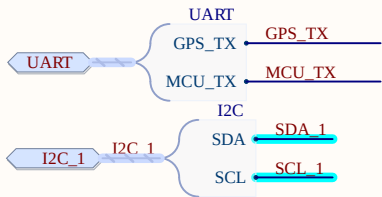


Digikey Parts

Desig.	Part Name	Desig.	Part Name
J1	3 Pin Header	U4	CAN Transciever
J2	34 Pin Ampseal		
J4	2 Pin Header		
J5	2 Pin Header		
J6	2 Pin Header		
J8	2 Pin Header		
J9	2 Pin Header		
J11	8 Pin Header		
BT1	1056TR		

Rev.	Events, Revisions, Details	Date
V1.0	Design completed to requirements.	08/05/2027





Timepulse drives at VCC - .04, so 2.9V and 4mA.

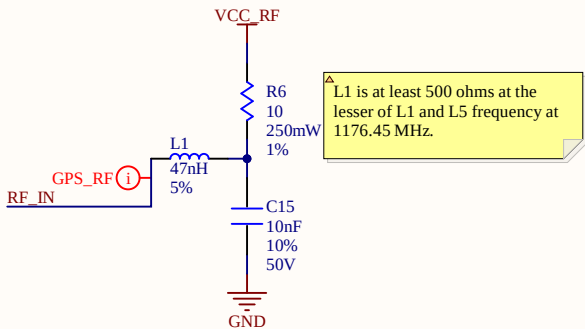
Resistor limits current to 3.3mA (when timepulse pulses):
DC Current gain is 60 (from datasheet using conservative value), so with 3.3mA we would therotically be able to drive a 0.198A load.
Diode should drive ~33mA, so well within specs.

C.1 Standard capacitors

Table 40 presents the recommended capacitor values for MAX-F10S.

Name	Use	Type / Value
C14	RF Bias-T capacitor	10 nF, 10%, 16 V, X7R

Table 40: Standard capacitors

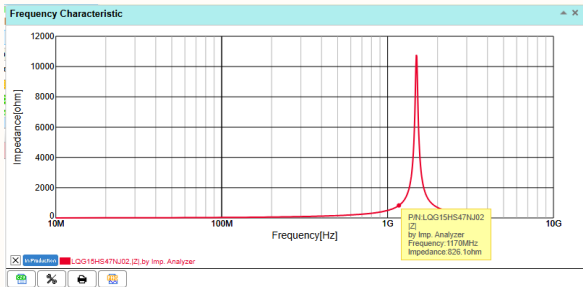


C.3 Inductors

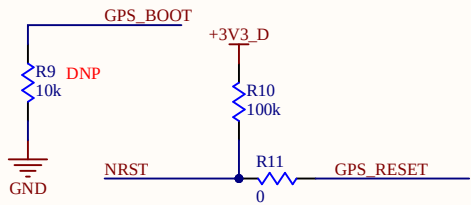
Table 42 presents the recommended inductor values for MAX-F10S.

Name	Use	Type / Value	Recommended component
L4	RF Bias-T inductor	47 nH, 5%	Murata LQG15HS47NJ02 Johanson Technology L-07W series Any other inductor with impedance > 500 Ω at GNSS L1 and L5 frequencies and current rating above 300 mA.

Table 42: Recommended inductors

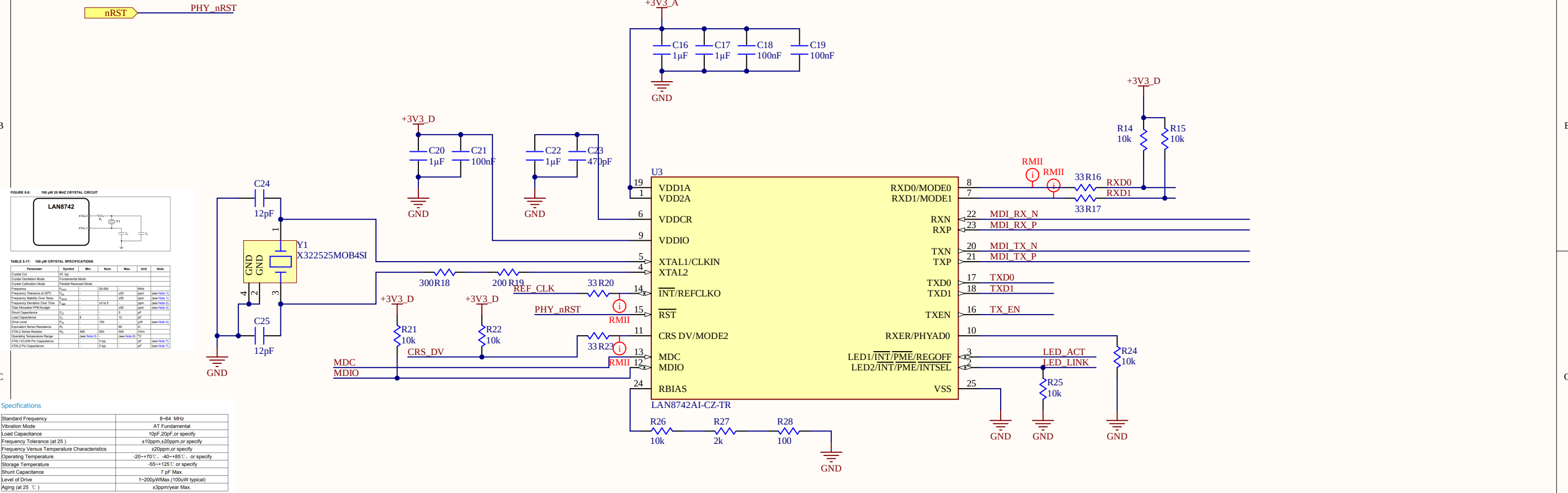
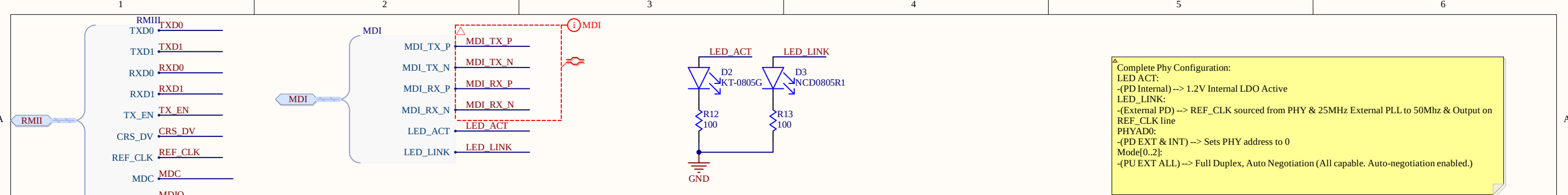


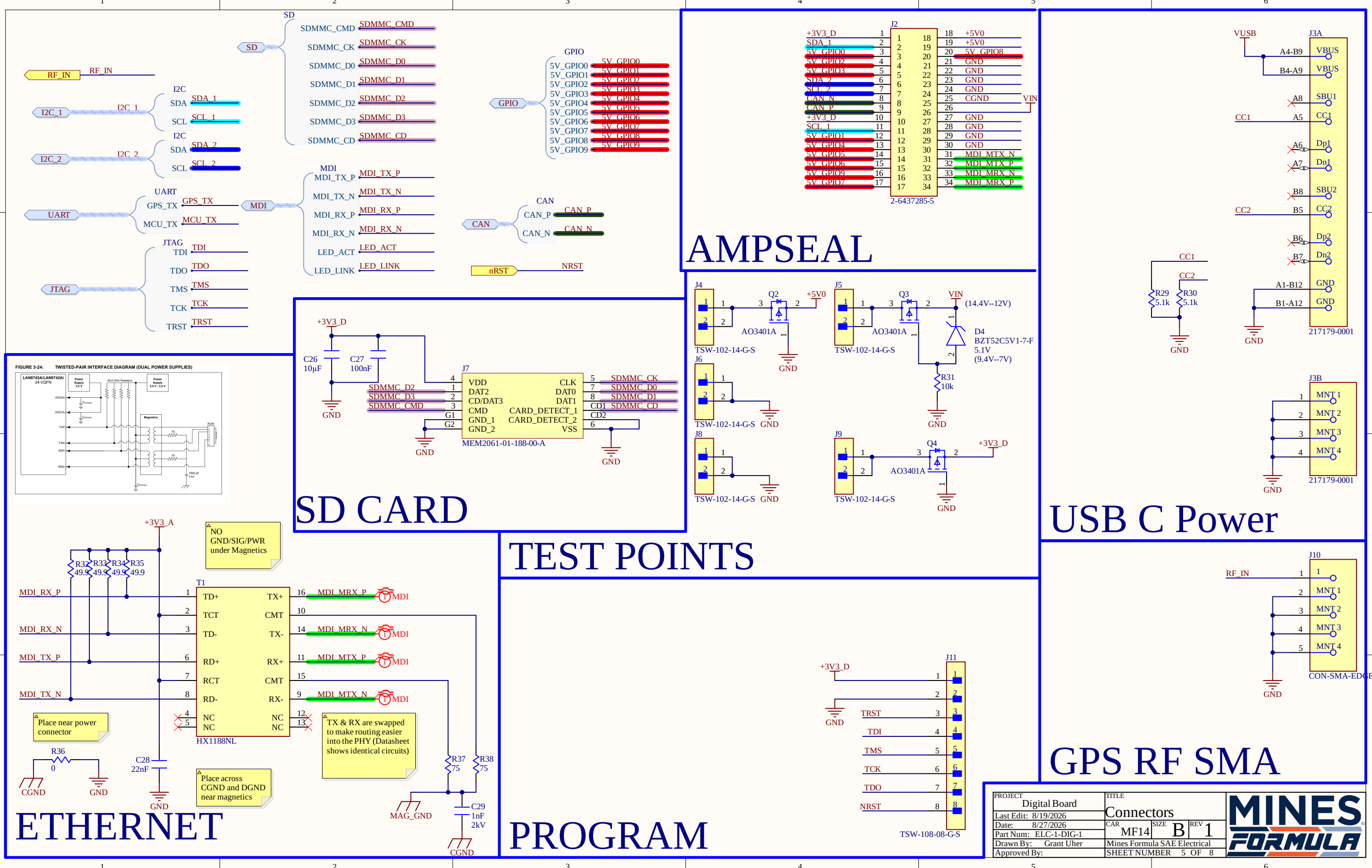
15	VIO_SEL	-	I	Voltage selector for V_IO supply. Connect to GND for 1.8 V supply, or leave open for 3.3 V supply.
16	ENA	2	IO	IO data bus enable (active low)
18	SAFEBOOT_N	-	I	Safeboot mode (active low). Leave open if not used. ¹⁴



CONFIG

PROJECT	Digital Board	TITLE			MINES FORMULA
Last Edit:	8/19/2026	GPS			
Date:	8/27/2026	CAR	SIZE	REV	
Part Num:	ELC-1-DIG-1	MF14	B	1	
Drawn By:	Grant Uher	Mines Formula SAE Electrical			
Approved By:		SHEET NUMBER 3 OF 8			





AMPSEAL

SD CARD

TEST POINTS

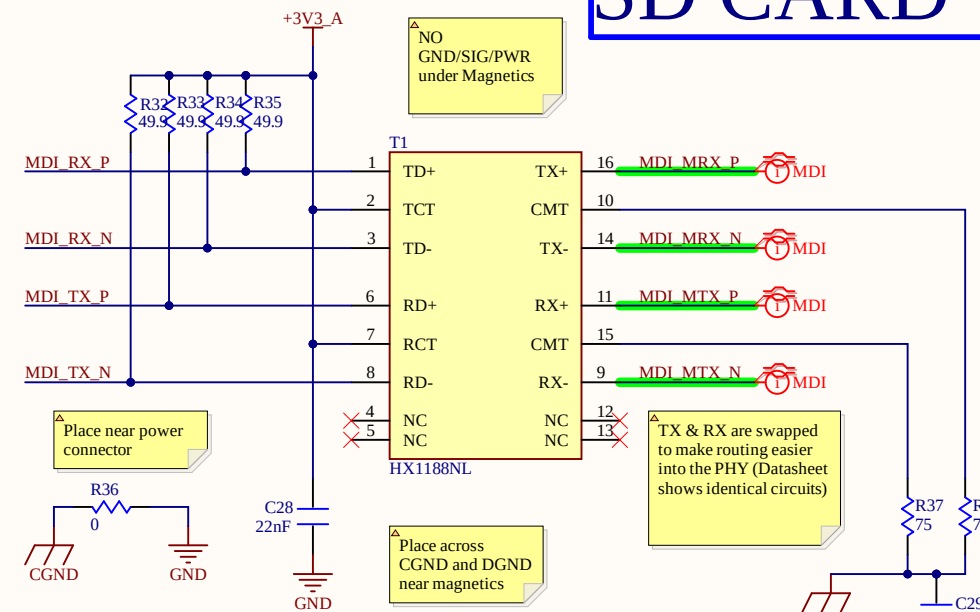
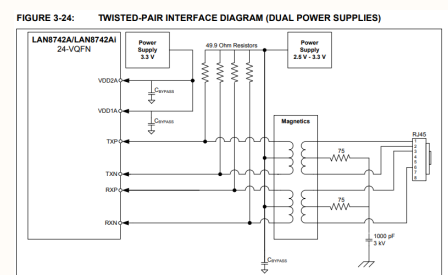
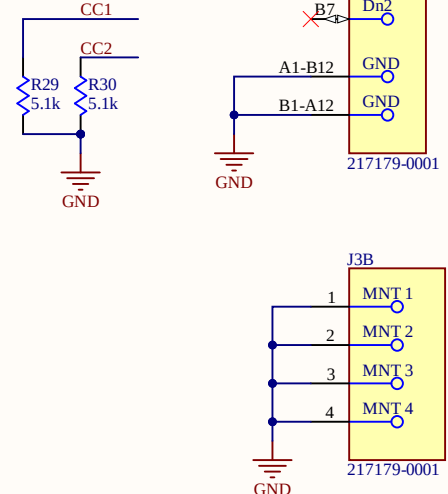
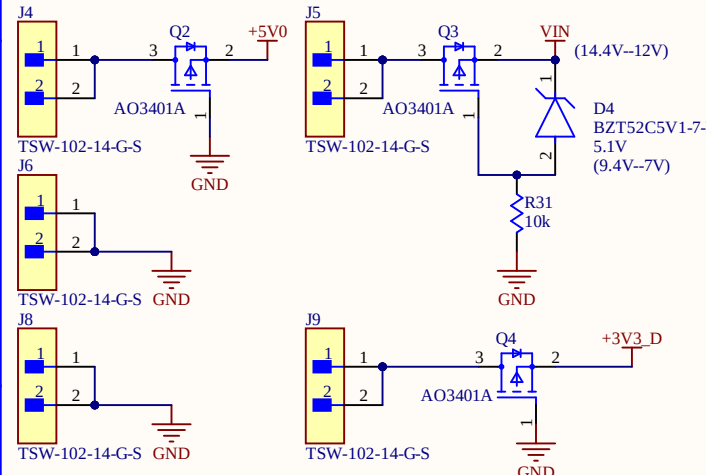
USB C Power

GPS RF SMA

ETHERNET

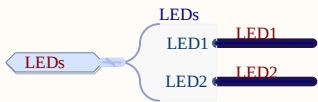
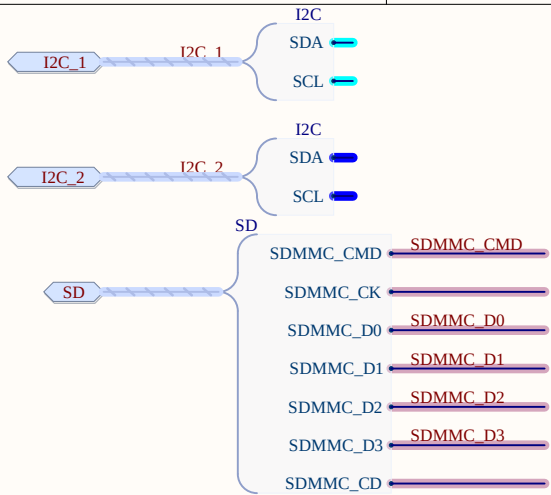
PROGRAM

J2	
+3V3_D	1
SDA_1	2
5V_GPIO0	3
5V_GPIO2	4
5V_GPIO3	5
SDA_2	6
SCL_2	7
CAN_P	8
CAN_N	9
+3V3_D	10
SCL_1	11
5V_GPIO1	12
5V_GPIO4	13
5V_GPIO5	14
5V_GPIO6	15
5V_GPIO7	16
5V_GPIO9	17
18	18
19	19
20	20
21	21
22	22
23	23
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25	25
26	26
27	27
28	28
29	29
30	30
31	31
32	32
33	33
34	34

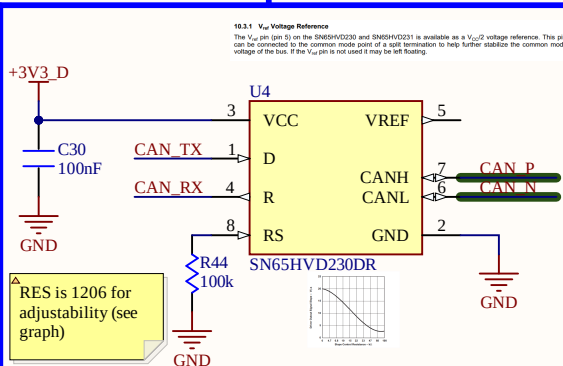
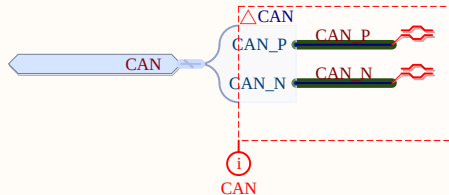
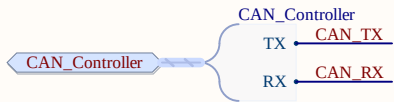
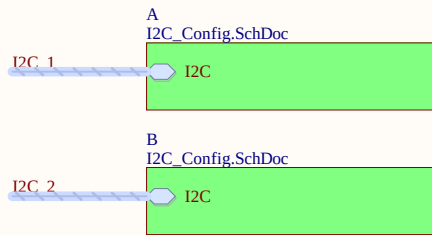


J11	
+3V3_D	1
GND	2
TRST	3
TDI	4
TMS	5
TCK	6
TDO	7
NRST	8

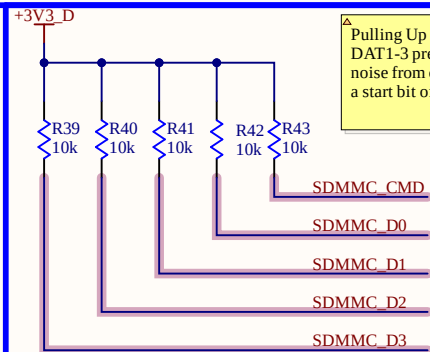
PROJECT	Digital Board			TITLE	Connectors		
Last Edit:	8/19/2026			CAR	MF14	SIZE	B
Date:	8/27/2026			REV	1		
Part Num:	ELC-1-DIG-1			SHEET NUMBER	5 OF 8		
Drawn By:	Grant Uher			MINES FORMULA SAE Electrical			
Approved By:				MINES FORMULA			



I2C

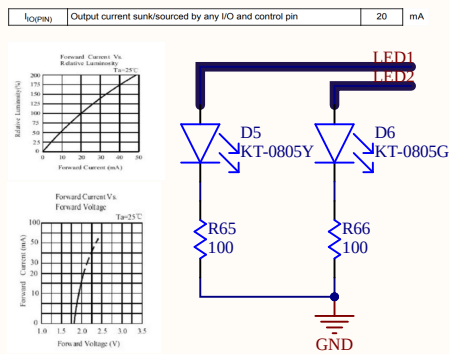


CAN DNP



SD Pull Ups

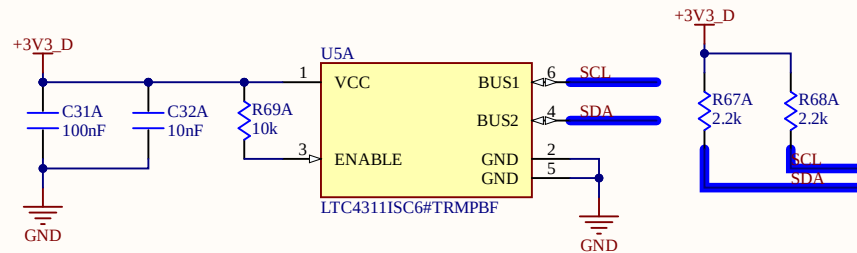
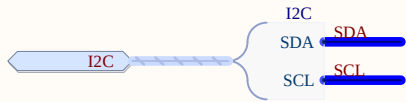
Pulling Up DAT1-3 prevents noise from calling a start bit of low.



LEDs

PROJECT	Digital Board			TITLE	Interface		
Last Edit:	8/19/2026			Date:	8/27/2026		
Part Num:	ELC-1-DIG-1			CAR	MF14	SIZE	A
Drawn By:	Grant Uher			Mines Formula SAE Electrical	REV	1	
Approved By:				SHEET NUMBER	6 OF 8		

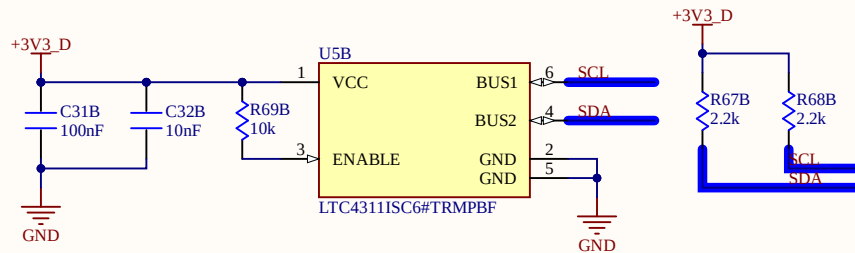
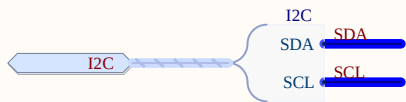




Place 10nF closer than 100nF caps

PROJECT	Digital Board			TITLE	I2C Range Extender			
Last Edit:	8/19/2026			CAR	MF14	SIZE	A	
Date:	8/27/2026			REV	1			
Part Num:	ELC-1-DIG-1			Mines Formula SAE Electrical				
Drawn By:	Grant Uher			SHEET NUMBER 7 OF 8				
Approved By:								





Place 10nF closer than 100nF caps

PROJECT	Digital Board			TITLE	I2C Range Extender			
Last Edit:	8/19/2026			CAR	MF14	SIZE	A	
Date:	8/27/2026			REV	1			
Part Num:	ELC-1-DIG-1			Mines Formula SAE Electrical				
Drawn By:	Grant Uher			SHEET NUMBER 7 OF 8				
Approved By:								

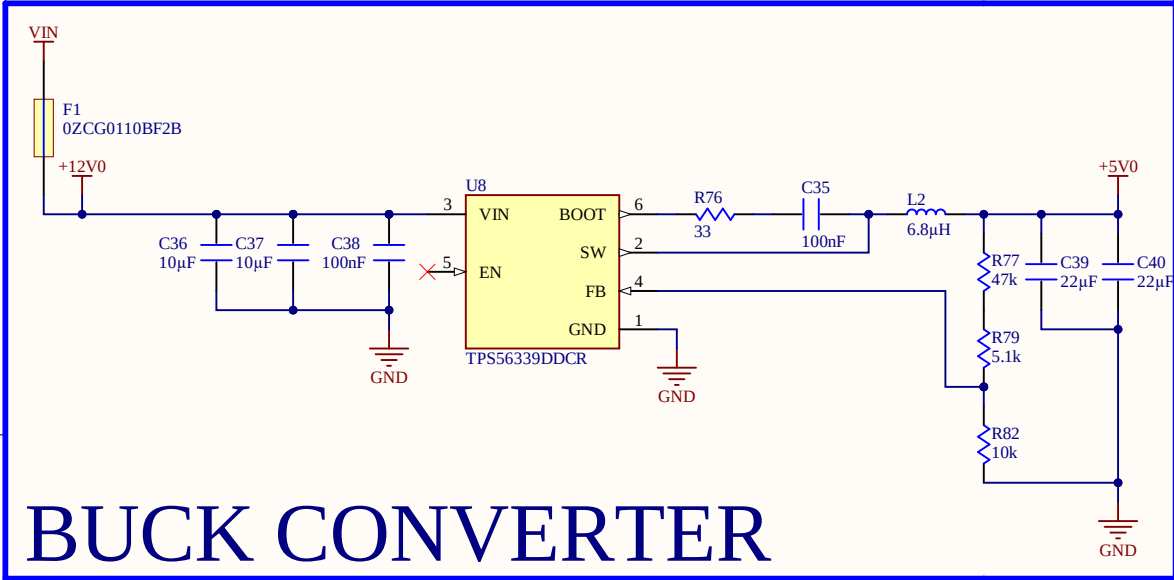


Power Budget:
12V:
LED: 12mA

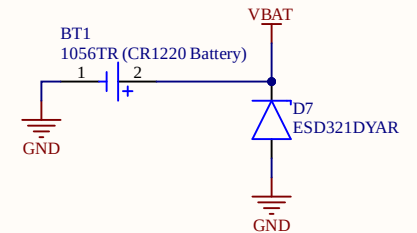
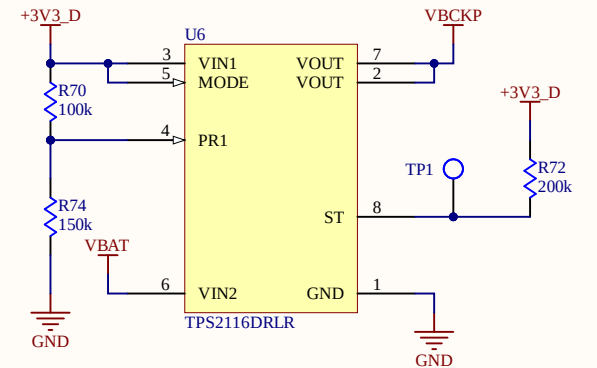
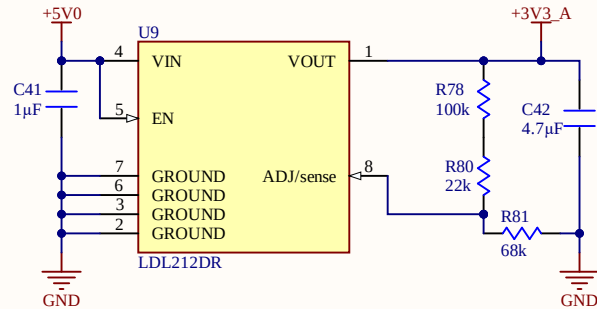
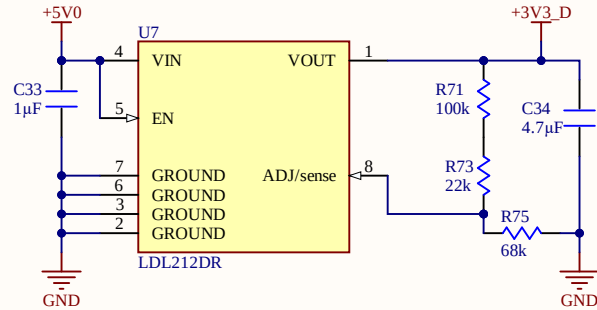
5V:
LED: 18mA

3.3V:
STM32: 120mA
NEO-M9N: 100mA
LAN8742A: 140mA
LEDs x2: 33mA

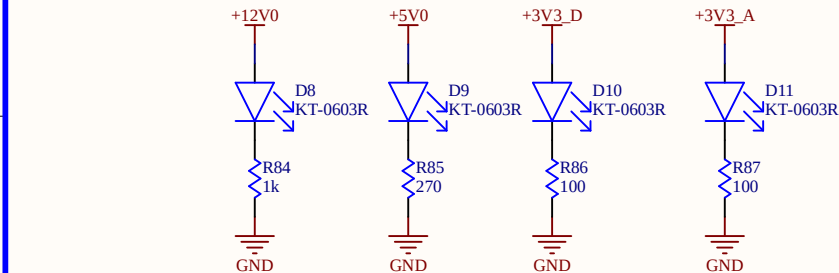
Total: around 450mA



BUCK CONVERTER

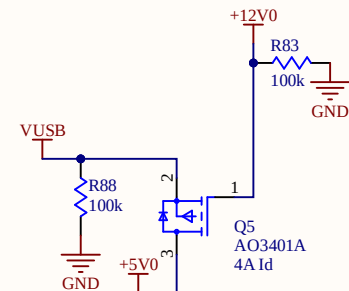


GPS BACKUP



LEDs

LDOs



USB Backpower Prot

PROJECT	Digital Board			TITLE			
Last Edit:	8/19/2026			Power			
Date:	8/27/2026			CAR	SIZE	REV	
Part Num:	ELC-1-DIG-1			MF14	B	1	
Drawn By:	Grant Uher			Mines Formula SAE Electrical			
Approved By:				SHEET NUMBER 8 OF 8			